

Neutron-2: A Dual CubeSat Mission for Three-Dimensional Mapping of Neutron Radiation in Low Earth Orbit

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ABSTRACT

Neutron interactions in Low Earth Orbit (LEO) remain a poorly characterized component of the space radiation environment, despite their potential to significantly impact spacecraft electronics, materials, and crew health. Unlike primary charged particles, most neutrons in LEO are secondary products generated when high-energy protons and heavy ions from galactic cosmic rays or solar energetic particle events interact with atoms and molecules in Earth's atmosphere. The production rate of these secondary neutrons is strongly dependent on atmospheric density, which varies with altitude, geomagnetic position, and solar activity, leading to spatial and temporal variability that is not well captured by current radiation models. The Neutron-2 CubeSat mission is designed to address this gap by measuring neutron and gamma-ray fluxes as a function of time, geomagnetic position, including altitude, latitude, longitude, and solar activity. By directly characterizing how neutron flux varies with orbital parameters, Neutron-2 will improve understanding of secondary neutron generation in LEO and contribute to more accurate radiation risk assessments for spacecraft systems and human missions. To enable these measurements, Neutron-2 employs a dual-spacecraft architecture consisting of two identical CubeSats deployed into distinct polar orbits at different altitudes. Operating concurrently, the two spacecraft will collect synchronized radiation measurements, enabling direct cross-comparison of neutron fluxes as a function of altitude while minimizing temporal and solar variability effects. The resulting dataset will support the generation of three-dimensional maps of neutron radiation in Earth orbit, providing new insights into the structure of the LEO neutron environment. Beyond its immediate scientific objectives, Neutron-2 serves as a technology and mission-operations pathfinder for future distributed radiation-monitoring missions. The dual-spacecraft configuration enables experiments in relative attitude control, formation awareness, and coordinated operations, laying the groundwork for more complex multi-spacecraft architectures. These capabilities directly inform a future mission concept, envisioned as a small-satellite swarm designed to measure radiation environments in cislunar space, where exposure levels and particle populations differ substantially from LEO and are increasingly relevant to sustained human exploration of the Moon and beyond. The neutron detectors for Neutron-2 are developed by Radiation Monitoring Devices, Inc. (RMD) in collaboration with Arizona State University (ASU), incorporating lessons learned from the Neutron-1 CubeSat mission launched in 2020. Neutron-2 introduces a next-generation payload and a new spacecraft bus designed to provide increased radiation tolerance and an extended orbital lifetime. The mission is led by students at the University of Hawai'i at Manoa (UHM), with mentorship from faculty and industry partners, and is supported by the Air Force Research Lab-sponsored University Nanosatellite Program (UNP). Neutron-2 advances scientific understanding of space radiation, matures distributed CubeSat mission operations, and trains the next generation of aerospace engineers for future multi-spacecraft missions.

INTRODUCTION

The Neutron-2 mission aims to fly two identical 3U CubeSats in a polar orbit to study neutron flux in Low Earth Orbit (LEO) as a function of time, geomagnetic position, and solar activity. By gathering neutron flux data related to these phenomena, a better understanding of the radiation environment can be attained, improving space weather models and methods to protect space assets from the hazardous events.

The mission is being developed by students at the University of Hawai'i at Manoa (UHM) as part of the University Nanosatellite Program (UNP). The UNP program funds university CubeSat projects to establish a strong aerospace and systems engineering workforce and give students hands-on experience developing a satellite mission¹. As part of the program, the student team has gone through the Preliminary Design Review and is preparing for the Critical Design Review in Fall of 2026.

SCIENCE OBJECTIVES & EXPERIMENT PLAN

Neutron interactions in the Low Earth Orbit environment remain under-characterized², and with this knowledge gap, the risk of damage to spacecraft and humans from space weather is high. The particles that generate neutrons have three different sources: 1) particles confined within Earth's magnetic field, 2) energetic solar charged particles that reach the Earth, and 3) high-energy protons and heavy ions, so-called galactic cosmic rays³. All three of these neutron generation sources pose a severe risk to spacecraft and humans as the particles can interact with matter and cause significant damage³. Without a full understanding of this environment, it is unclear how to protect the assets invested in space and prevent this damage.

There are three distinct phenomena that the Neutron-2 mission is interested in. The first of which is how Earth's Magnetic Field affects neutrons. The magnetic field around Earth is affected by the Lorentz force, which accounts for the radial acceleration of charged particles. They curve perpendicular to their initial movement vectors, $\mathbf{v}$, and magnetic field, $\mathbf{B}$. A larger angle between $\mathbf{v}$ and $\mathbf{B}$ results in a stronger deflecting force. A particle's speed also affects its deflection. The slower the particle, the lower its kinetic energy, and the greater its deflection from its initial path. Near the equator, particles travel perpendicular to the magnetic field, and near the poles, particles travel parallel to the magnetic field⁴. This path of deflection is a way to visualize the shielding effects that the magnetic field has on Earth. By gathering neutron data at different

magnetic latitude points, it can be better understood how this shielding phenomenon works.

The second phenomena that the mission is interested in is how altitude, or the depth of Earth's atmosphere, affects neutron counts. The majority of neutrons and gamma-rays that will be observed in LEO will be secondary particles that are produced by interactions between galactic cosmic rays (GCR) or solar energetic protons and atmospheric molecules. Upon entering the atmosphere, the particles will interact with the atomic nuclei of atmospheric nitrogen and/or oxygen and secondary neutrons create a nuclear particle cascade in the atmosphere until their energy is entirely consumed by nuclear interactions. This leads to the build-up of an inter-nuclear cascade as shown in Figure 1 leading to increased neutron fluxes^{4,5}. The development of the particle cascade indicates that atmospheric depth will have an influence on the number of neutrons occurring in each measured region. As measurements are taken throughout the atmosphere, there is an increase in secondary neutrons as the cascade approaches Earth.

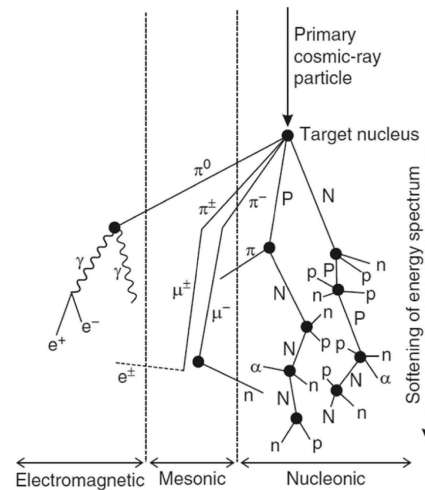


Figure 1: Secondary cosmic ray cascade in the atmosphere. Abbr: n, neutron, p, proton (capital for particles carrying nuclear cascade), α , alpha particle, e, electron or positron, γ , gamma ray, π , pion, μ , muon¹.

The third phenomenon that the mission is interested in is how solar events affect the production of secondary neutrons and gamma-rays. Aside from GCRs, particles from solar events shall also yield secondary neutrons when interacting with particles in Earth's atmosphere. Therefore, when a solar event occurs, it is expected that there would be a jump in neutron counts. By monitoring when events occur and taking data before, during, and after them, a better understanding of the effects can be made. Additionally, the 11-year solar cycle and its effects on neutron counts will also be monitored as the

flux of GCR's is modulated by variations of the solar magnetic field.

Based on communications with the stakeholders and an understanding of the science phenomena that are of interest for this mission, the following mission objectives and success criteria were defined:

Mission Objectives (MO):

MO1. Measure the flux of secondary neutrons and gamma rays produced by cosmic ray interactions in Low Earth Orbit as a function of time, geomagnetic latitude, and altitude (location based data).

MO2. Measure secondary neutron and gamma ray flux variations before, during, and after solar energetic particle (SEP) events (event based data).

Success Criteria (SC):

Success criteria have been derived from each MO, with a minimum success criteria (MSC) being the minimum needed for the mission to be deemed successful, and full success criteria (FSC) being the goal of the mission. Each success criteria is listed below with the number correlating to the MO that they are a criterion for. For MO-1, an extended success criteria (ESC) was defined to capture the goal of having two satellites.

MSC1. The mission shall collect at least 300 total neutron counts distributed uniformly over one of the Earth's poles (see definition 7). To ensure uniform collections, the region shall be subdivided into three equal latitude cone sections (see definition 13). The accumulated neutron count within each cone section must have a maximum relative uncertainty of 10% of the neutron flux model (from Poisson's statistics ~ 100 neutron counts per cone section).

MSC2. Collect neutron and gamma ray flux measurements before, during, and after a single solar energetic particle (SEP) event within one of the regions of interest (see definition 9). The before and after ("baseline") measurements used for comparison shall occur no more than 40 days before and after the SEP event within the same region. At least one baseline measurement needs to occur between two sequential SEP events to validate the event observation(s).

FSC1. The mission shall collect neutron flux measurements in each of the following regions: the North Polar region (latitude $>66^\circ\text{N}$), the South Polar region (latitude $<-66^\circ\text{S}$), and the South Atlantic Anomaly (SAA), with sufficient counts to achieve $\leq 1\%$ statistical uncertainty of the neutron flux model in each

region (from Poisson's statistics $\sim 10,000$ counts per region, or 30,000 total).

To enable temporal comparison of the neutron environment in Low Earth Orbit, the mission shall collect an additional dataset after six months in each region achieving the same statistical uncertainty of the neutron flux model.

FSC2. Collect neutron and gamma ray flux measurements before, during and after solar energetic particle (SEP) event(s) within each area of interest (NP, SP, and SAA). For each SEP event, the before and after ("baseline") measurements used for comparison shall occur no more than 40 days before and after the SEP event within the same region. At least one baseline measurement needs to occur between two sequential SEP events to validate the event observation(s).

ESC1. The mission shall complete FSC-1 at two orbital altitudes separated by at least 50 km (12 datasets total). For each region, the two separate altitude datasets shall be obtained within 40 days of each other. This enables comparison of neutron flux as a function of altitude while minimizing the influence of solar cycle variation.

Payload Design

The neutron detector that is being used for this mission is designed and developed by Arizona State University and Radiation Monitoring Devices. It is an updated version of the detector that has flown on the Neutron-1⁶ and LunaH-Map^{7,8,9} missions, and although the electronics, housing, and scintillator material have been updated, the principles remain the same.

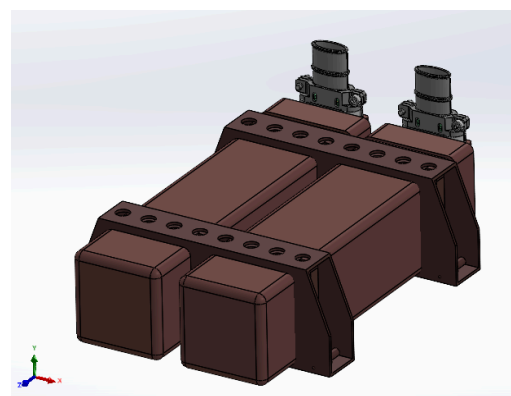


Figure 2: CAD Model of the N2 payload modules, one wrapped in Gd and one wrapped in Tin.

The detector utilizes two sensor heads, each made of a novel scintillator single material called TL₂Y₂C₁₆, which enables the simultaneous acquisition of gamma ray and neutron spectroscopy. Gadolinium (Gd) can absorb thermal neutrons below 0.3 eV but not

as strongly absorb neutrons above 0.3 eV (epithermal). By having two sensor heads, one wrapped in Gd, one not, the payload can then differentiate between epithermal and thermal neutrons. The two sensor heads will each have their own photomultiplier tube (PMT) but share a payload computer for processing and storage of the data. The detector is modular in design to minimize development risk, as seen in Figure 2 and Figure 3.

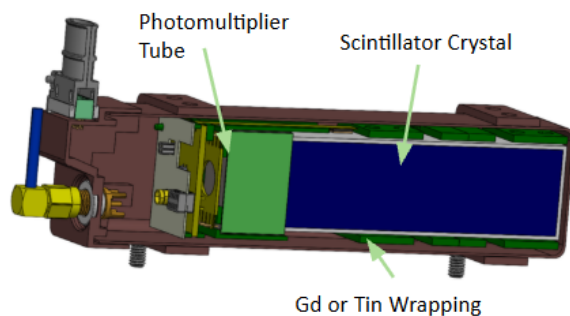


Figure 3: CAD Model of the interior of a N2 payload module.

The instrument is sensitive to all types of radiation, and as particles hit the TYCL scintillator, a distinct light pulse is generated for different types of radiation. Neutrons and gamma rays are able to be isolated using pulse shape analysis techniques, specifically, the rise time is used to distinguish between the two. The instrument can then determine neutron and gamma-ray flux throughout LEO by determining changes in the number of neutrons per square centimeter per second.

There are two primary operational states for the payload: Standby State and Data Acquisition State. While in the former, the payload parameters can be configured and the science data can be transferred to the satellite OBC. When commanded into Data Acquisition State, the payload will collect particle count data and store it locally. This state is set for a specific duration, once reached, it returns to Standby State and processes the data.

Experiment Plan

To fulfill the mission objectives and success criteria, three primary experiments have been defined to complete when on orbit. The first of which is related to Mission Objective 1 and aims to collect science data as a function of the solar cycle and geomagnetic latitude. Experiments 2 and 3 are related to Mission Objective 2 with a goal of collecting science data before, during,

and after solar events. Experiment 2 aims to collect science data during solar events, whereas Experiment 3 will collect baseline data (before and after solar events) in the event that baseline data is not collected from Exp. 1.

Experiment 1 will focus on collecting at least 10,000 neutron counts within each of the regions of interest: North Pole, South Pole, and South Atlantic Anomaly (SAA). To complete this, the operators will focus on collecting data one region at a time until the necessary counts are reached, and then move on to another region. Collections will be done in the polar regions first as those regions fall higher in priority than the SAA region based on customers' needs and data gaps. The 10,000 counts within each region will need to be collected within a 40-day period to limit the effect of the solar cycle as 40 days is approximately a 1% change in the cycle. Collecting within 40 days will allow the customers to isolate the results from the magnetic field and not the solar cycle. Since the solar cycle is also of interest, at least 6 months after the initial 10,000 counts are taken in each region, another collection of 10,000 counts should be completed again to understand how the change in solar cycle has affected the neutron counts.

For Experiment 2, mission operators will be monitoring space weather databases for solar events. If an event is detected, data collection will be commanded to acquire data within the next collection window during the event. Typically, Experiment 1 will be occurring at all times in a specific region of interest. If a solar event is detected, Experiment 2 will focus on collecting science data in the region of interest that Exp. 1 is already focussing on. This will allow the data from Exp. 1 to be used for the baseline measurements for before and after the event.

Baseline measurements before and after the solar events will need to be collected within 40 days of the actual event to prevent changes from the solar cycle affecting the results. Figure 4 shows an example sequence of events for collecting data before, during, and after solar events. To attain these baseline measurements, the experiments are being designed so that the data collected during Exp. 1 can be used as the baseline. However, if Exp. 1 data fails to provide this baseline data, Experiment 3 exists as a Baseline Data Collection experiment to collect any necessary data not captured by the other experiments.

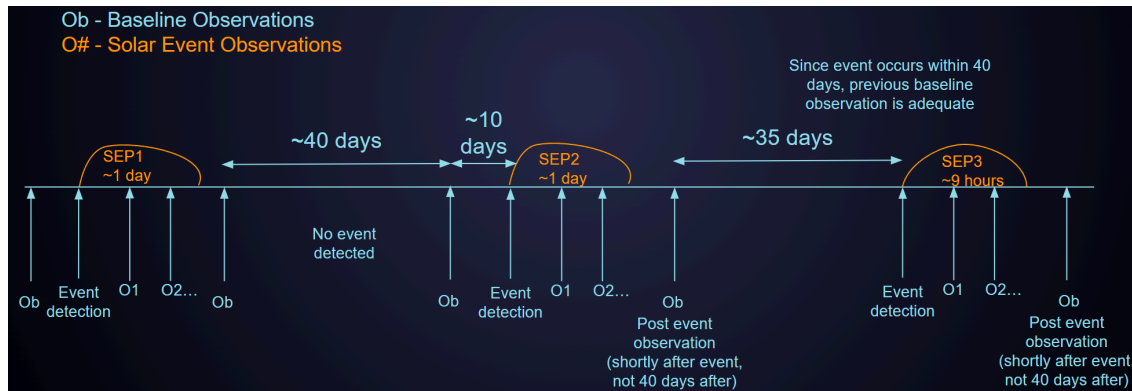


Figure 4: Baseline event sequence for Experiment 2. Baseline measurements should occur before and after an SEP. If no SEP is detected within a 40-day window, reset the clock and run another set of baseline observations.

MISSION DESIGN

To fulfill the mission objectives, the team aims to build and launch two 3U CubeSats into polar orbits with a difference of altitude of at least 50 km. In order to achieve this and reduce complexity, the team plans to design only one satellite and plan to build two copies. These two identical satellites, N2-A and N2-B, will then be on the same launch and deployed at different times in order to achieve the desired altitude difference. Initially the team looked into using methods such as drag, propulsion, or different launches to achieve the altitude difference. However, using methods such as drag or propulsion meant that the two satellites would have to have a different design and operations plan, increasing the complexity of the design, build and operations phases beyond the scope of a student project. Different launches was not a probable option as SmallSats are secondary payload on launch vehicles and having two launches into the correct orbits within the same time period was not likely. Therefore the team plans to only design one satellite, build two copies of it, and during operations have the satellites each complete the three experiments individually. The data can then be compared on the ground to understand how altitude affects neutrons.

Due to the high risk of getting a second satellite fully functioning in orbit, with risk sources being related to lack of resources (personnel, funding, etc.), difficulty commissioning the satellite on orbit, and launch options, the success criteria have been defined to allow the team to achieve minimum and full success with only one satellite. Extended success criteria 1 exist to capture the goal of understanding how altitude affects the neutron environment in LEO.

Overall System Architecture

The design for the satellite system can be seen in Figure 5. The system is made up of the Onboard Computer Subsystem (OBC), Communications Subsystem (COMM), Electrical Power Subsystem (EPS), Attitude Determination and Control Subsystem (ADCS), Thermal Control Subsystem (TCS), Structures and Mechanisms Subsystem (STR), Flight Software Subsystem (FSW), and the Payload Subsystem (PAY).

The bus, mostly made up of Commercial Off the Shelf (COTS) components, uses 3.3 Volts, 5 Volts, and Battery Voltage (VBatt) all supplied from the EPS. I2C is the primary method of communications between the subsystem, with SPI being used to communicate between the Radio and OBC. The following section goes deeper into each subsystem and their design.

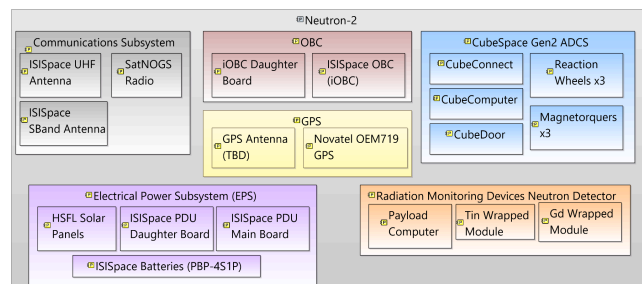


Figure 5: Neutron-2 System Level Block Diagram including varying subsystems: Communications (gray), OBC (red), ADCS (blue, with GPS in yellow), EPS (pink), Payload (orange).

SATELLITE BUS

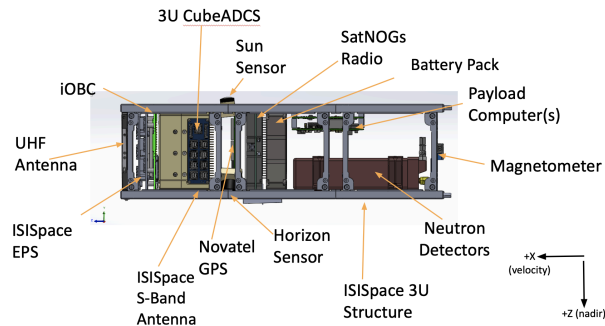
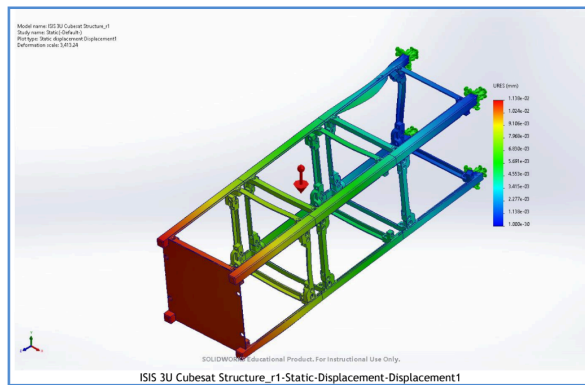


Figure 6: CAD model of satellite bus design with main subsystems bay on the left and payload bay on the right.

Structures and Mechanisms

The Neutron-2 bus will use a COTS 3U form factor structure from ISISpace¹⁰. This 3U configuration allows the mission to be completed while keeping the satellite within the FCC five-year deorbit requirement. The decision to use two 3U buses was driven by the mission's interest in orbital constellations and formation flying. A 3U platform also provides an efficient balance between available volume and development cost.



Name	Type	Min	Max
Strain1	ESTRN: Equivalent Strain	1.771e-10 Element: 32	7.553e-04 Element: 422

Figure 7: Initial Finite Element Analysis of N2 Structure. Based on the material properties of Aluminum 6061-T6 and the predicted load during launch, the ultimate Factor of Safety (FoS) is calculated to be 2.97. The typical ultimate FoS required from launch providers is 1.40.

The ISISpace structure includes two built-in inhibit kill switches that keep the satellite powered off until deployment. This ensures that no power or current flows through the satellite during launch and early orbit operations. Initial finite element analysis (FEA) testing, as seen in Figure 7, showed that the structure is

expected to meet the mission's structural integrity verification requirements. In addition, because of the flight heritage of this structure, it is considered a reliable and flight-proven option for launch.

Thermal Design

The N2 thermal analysis is still in a preliminary stage, but the model is becoming more defined as the design matures. Current component interface control documents (ICDs) show that each subsystem and component has its own operating and survival limits, as seen in Table 1. The thermal design must ensure that all components remain within their individual allowable ranges throughout the mission.

Table 1: N2 Operating (green) and Survival (Orange) Temperature Limits.

Temperatures in Degrees Celsius		-60	-50	-40	-30	-20	-10	0	10	20	30	40	50	60	70	80	90
Component																	
Structure																	
Harnessing																	
Radio																	
UHF Antenna																	
S-Band Antenna																	
iOBC																	
ADCS																	
GPS																	
Payload																	
Solar Panels																	
Battery Pack																	
PDU																	

Initial thermal modeling began with multi-node hot-case and cold-case scenarios. In the hot case, the satellite spends a greater portion of the orbit in sunlight, represented here by a Right Ascension of the Ascending Node (RAAN) of 90°. In the cold case, the satellite spends a greater portion of the orbit in eclipse, represented by a RAAN of 0°.

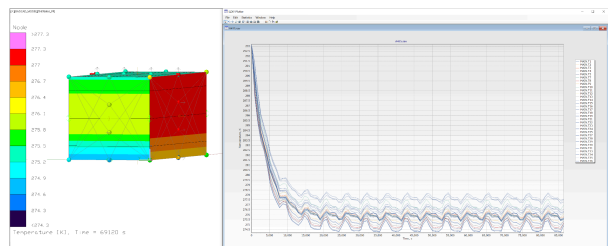


Figure 8: The Hot Case of Thermal Modeling with gradient shown. The number of Nodes is 36.

Preliminary modeling currently predicts orbital temperature ranges of 5.5 °C to 1.35 °C during the hot case and -15.15 °C to -32.15 °C during the cold case. These values will be refined as the model is updated. As the design continues to close, the thermal model is being improved by replacing simplified black-box thermal loads with component-level thermal loading for each subsystem. The current orbit is too cold for the satellite to operate nominally in which we may include heaters as one of our components during flight. Developing a higher-fidelity model will provide a better understanding of which components may require thermal control features so that the spacecraft can remain within acceptable operating temperatures throughout orbit. Temperature sensors are also being considered for implementation at the subsystem level.

Electrical Power Subsystem

The N2 satellite will use the ISISpace Compact Electrical Power Subsystem (iCEPS)¹¹, which provides clean and reliable connections to the iOBC through an I2C interface. The EPS includes a 45 Wh battery and multiple power outputs that support 3.3 V, 5 V, and a programmable VBatt line which supports up to 32.1 V¹¹. These outputs can be used to power the rest of the spacecraft bus as well as the payload.

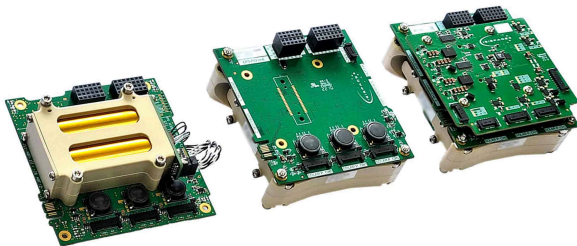


Figure 9: ISISpace iCEPS Type A, B, and C.

For the N2 Engineering Model (EM), the Power Distribution Unit (PDU) from the Artemis Cubesat Kit will be used, which was developed at the University of Hawai'i in 2020. The PDU has many of the same power interfaces as the ISISpace iCEPS and will also be a proof of concept of the flight system, as the OBC and the EPS system share the same manufacturer.

Attitude Determination and Control Subsystem

The N2 mission will use a COTS, turn-key attitude determination and control subsystem from CubeSpace¹⁵. HSFL has previously used CubeSpace Gen 1 components and is already familiar with both their software and the Dawn Dusk Satellite Simulator (D2S2, <https://dawn dusk.space>), which supports attitude control testing and integration with the iOBC.

HSFL can also perform live testing with the Attitude Control Testbed built in 2014¹⁶. By connecting a unit to D2S2, the team can demonstrate the same command paths used by a flight model, providing an engineering model proof of concept.

The 3U Gen2 ADCS from CubeSpace is an updated version of their first-generation unit. The configuration used for this mission will consist of 3 reaction wheels, 3 magnetorquers, 2 Fine Sun Sensors, 1 Nadir Sensor, 6 Coarse Sun Sensors, 2 magnetometers, 1 Gyro Rate Sensor, and an ADCS Computer. With this combination of actuators and sensors, the ADCS is capable of achieving the 2° of freedom pointing required for the S-Band antenna.

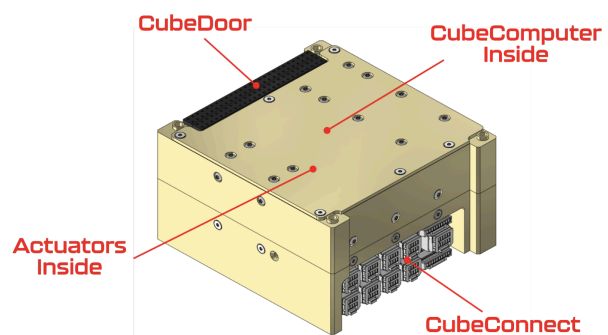


Figure 10: CubeSpace Gen2 3U Core Component ADCS

Communications

The N2 mission will use the SatNOGS COMMS radio board, a half-duplex, turn-key solution capable of both UHF and S-band communication.

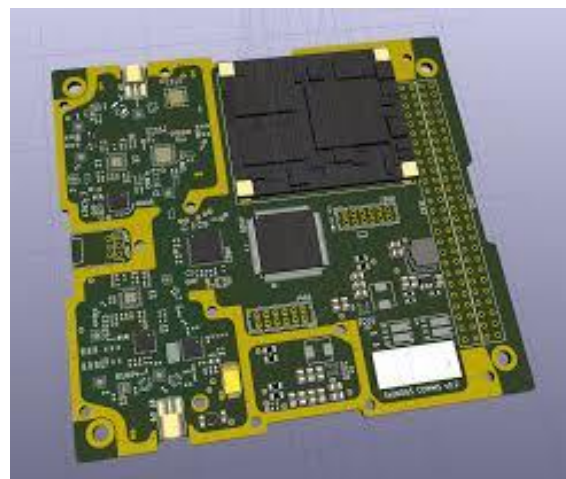


Figure 11: SatNOGS COMMS Radio Board provides UHF up/down and S-band up/down links.

The radio was designed by the Libre Space Foundation and is an open-source system with schematics and documentation available through its GitHub repository¹⁷. SatNOGS is also supported by an active engineering community that is continuing to develop the board and build its flight heritage by supporting the 0.3.2 PHASMA production.

The dual-band communication architecture allows the mission to use UHF for state-of-health telemetry and command uplink, while S-band is used to downlink science data due to its larger data volume. The mission will use the ISISpace 3U deployable antenna system¹² and the ISISpace S-band patch antenna¹³. Both components are commercially available and have demonstrated reliability and performance specifications that align with the mission requirements. The SatNOGS boards will be assembled and manufactured by students at UHM. Initial production will focus on engineering model boards, which will serve as a proof of concept for a completed design review and a demonstration of the mission life cycle. Once the design is validated at the engineering model level, flight model boards will be procured from a qualified vendor.

On-Board Computer Subsystem

The iOBC¹⁴ from ISISpace will serve as the primary flight computer for the N2 mission. HSFL has previously used the iOBC CubeSat missions due to their reliability and the space heritage of the company. The iOBC provides multiple data interfaces, allowing the team to integrate effectively with the spacecraft components. For the engineering model flatsat, the team will use Raspberry Pi Zero W units as development-model OBCs to act as a functional equivalent to the iOBC.



Figure 12: ISISpace Onboard Computer

The Raspberry Pi similarly offers multiple data interfaces, including GPIO connections, and uses an ARM-based processor architecture comparable to that of the iOBC. This makes it a practical platform for maturing the flight software before flight hardware integration. Because of these similarities, the team

expects that transitioning from the development model to the flight model will require fewer software changes, reducing both integration time and overall effort.

Flight Software Subsystem

The team is currently developing an NASA JPL F' (F-prime) based flight software architecture for the mission. F' was developed by the Jet Propulsion Laboratory (JPL) and is an open-source framework that promotes community collaboration¹⁸. HSFL is also working toward developing in-house expertise with this architecture and aims to use F' across multiple missions at the university.

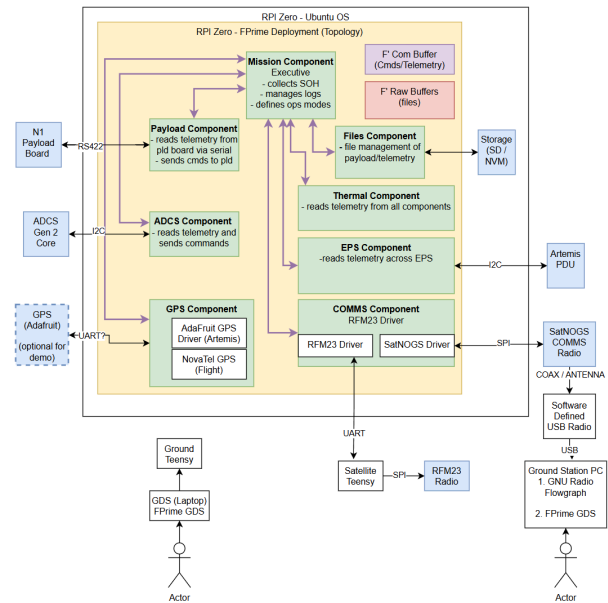


Figure 13: Flight Software Subsystem Architecture Diagram highlighting the different F' components for the FlatSat demo.

As the flight software is developed, the team is both contributing to and benefiting from a broader community of universities and teams that use the same framework. F' supports rapid development through its ability to generate system topologies within the codebase. As the team works through challenges such as limited software expertise, the F' community and architecture provide a strong foundation for developing a reliable flight software system for the mission.

FLATSAT

To verify the system design, the team is building a FlatSat composed of engineering model (EM) components. These components are being chosen to closely replicate the functionality of the Flight Model components while keeping the cost at a minimum. Table 2 shows components for both the FM and EM.

Subsystem	Flight Model Component	FlatSat (EM) Component	Difference between FM and FS (not on FM)
OBC	ISISpace iOBC	RPi 0W	The ISIS iOBC is flight-ready, radiation-tolerant, power-efficient, and real-time, while the Artemis OBC is cheaper, more flexible, faster to develop, but higher-power, less protected, and less flight-representative.
EPS	ISISpace iCEPS	Artemis EPS	iCEPS2 Type C is a commercial flight EPS with more power capability and protection. Artemis PDU/EPS is a simpler educational kit power system built for accessibility and development.
ADCS	CubeSpace ADCS Gen2 3-Axis CubeADCS	CubeSpace Gen 2 Developer Model (DM) Boards	CubeSpace DM: - No sensors or actuators (use D2S2)
PLD	N2 Payload Board	N2 Payload Board	The payload computers will be the same as the flight model.
COMMS	LibreSpace SatNOGs	LibreSpace SatNOGs	EM Components will not be space grade.
COMMS	ISISpace Deployable 3U Antenna	Artemis UHF Antenna	The ISISpace antenna is a commercial, qualified, flight-ready deployable UHF/VHF antenna system, while the Artemis measuring-tape antenna is a simpler, low-cost educational deployment mechanism mainly used for prototyping and teaching.
COMMS	ISISpace S-Band Patch Antenna	TBD	TBD
GPS	NovaTel GPS	TBD GPS	Artemis GPS
S&M	ISISpace 3U Structure	FlatSat Brackets	EM structure will likely be mainly made up of 3D printed parts.
TCS	Unspecified Thermal Connection	N/A	TBD

Table 2: Components for the Flight Model and the Engineering Model and their differences.

MISSION OPERATIONS

The Neutron-2 mission is divided into three operational phases: Launch and Early Orbit Operations (LEOP), Engineering Evaluation and Checkout (EE&CO), and Nominal Operations. LEOP begins immediately after orbital insertion and focuses on establishing communications with the spacecraft. Once communication is established, EE&CO begins, during which both the satellite bus and payload are commissioned to verify survival through launch and proper system functionality. After checkout is complete, the mission enters Nominal Operations, where science experiments are carried out. To further define spacecraft behavior during these phases, the mission uses a set of operational modes and submodes that describe specific spacecraft goals and the configurations required to achieve them.

For Experiment 1, the spacecraft begins in Base Mode with an attitude optimized for power generation. During ground contact, telemetry is downlinked, and future science collections or downlinks are scheduled using geolocation commands informed by orbit analysis tools. After contact ends, the satellite remains in Base Mode until a scheduled collection is triggered. At that point, it enters Science Collection Mode, transitions to an LVLH attitude, powers the payload into standby state, and then data acquisition state, where it can collect particle data. Data collection is expected to last approximately 15 minutes over the polar regions. Afterward, the spacecraft enters Science Processing Mode, where the payload processes the data and transfers it to the OBC

for later downlink, then returns to Base Mode. This cycle is repeated for future collections.

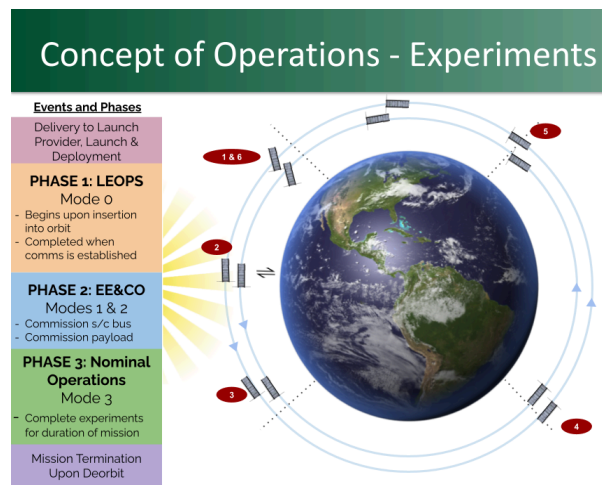


Figure 13: Concept of Operations demonstrating the steps for a nominal data collection for Experiment 1. The steps are as follows: (1.) Satellite in Nominal Base Mode, (2.) PASS over GS in Base Mode or Science Downlink Mode, (3.) GEOLOCATION command(s) triggered, enter Science Collection Mode for TBD minutes, (4.) DATA collection timer expires, enter Science Processing Mode for TBD minutes then Nominal Base Mode, (5.) GEOLOCATION command(s) triggered, enter Science Collection Mode for TBD minutes, (6.) DATA collection timer expires, enter Science Processing Mode for TBD minutes then Nominal Base Mode

Science data will preferably be downlinked in a dedicated Science Downlink Mode using the S-band

radio, since UHF alone would require significantly more passes to return the same volume of data. UHF will remain the preferred option for state-of-health telemetry because it requires less power and has fewer attitude constraints.

Experiments 2 and 3 follow this same general operational flow, with differences in when and where data is collected. Experiment 2 follows the same collection, processing, and downlink sequence, but data acquisition is driven by solar events rather than routine scheduling. Operators will monitor solar weather databases such as SpaceWeatherLive.com¹⁹ which pulls data from NASA's Solar Dynamic Observatory²⁰ to identify events and then schedule collections in the relevant regions, likely in the same region Experiment 1 is currently being completed in. Once an event is identified, the team will have one to two days to schedule data collection during the event, followed by a post-event collection over the same region for comparison. Pre-event baseline data may come from either Experiment 1 or 3 collections.

Both satellites, at their different altitudes, will complete the operations plan independently. The commands and data to and from the mission operations center will be tagged with a unique identifier, identifying whether the command and data is with respect to N2-A or N2-B. All science data will be compared on the ground and the two distinct satellites will not need to communicate with each other nor be aware of the other's location or state.

MODEL-BASED SYSTEM ENGINEERING IMPLEMENTATION

In addition to providing students with experience in mission development, the Neutron-2 mission is being used in Model-Based Systems Engineering (MBSE) research conducted at UHM. Research is underway to understand how MBSE can be used for student missions to improve efficiency and reduce mission costs. Specifically, Neutron-2 is being used as a case study to understand how MBSE methodologies can capture real mission information^{21,22}.

The Model-Based Systems Engineering approach for Neutron-2 was developed using the Arcadia method implemented in Capella, applying a SmallSat-tailored methodology that maps the project's life-cycle phases to Arcadia's architectural layers.

Operational Analysis was used to capture the scientific phenomena driving the mission, namely the secondary neutron and gamma-ray fluxes produced by cosmic-ray interactions and solar energetic particle events, ensuring that the underlying science needs were well understood

before solution development began. This was followed by a System Analysis layer capturing the Neutron-2 mission statement, objectives, and success criteria within a System Mission Capabilities Diagram, with derived and satisfied relationships established to support requirements traceability.

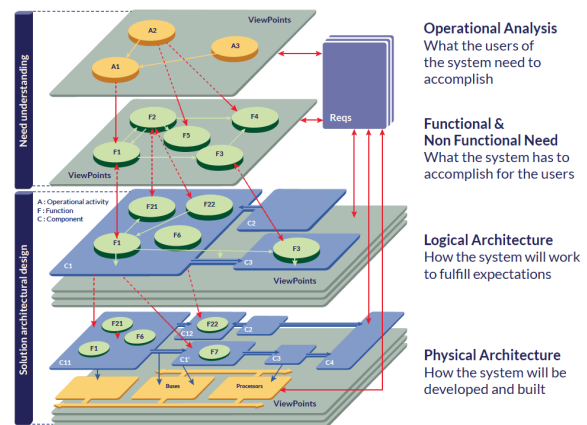


Figure 14: Description of the Arcadia method and process. Arcadia is a model-based engineering method.

The Logical Architecture layer was then used to define the functional behavior of the payload, modeling how the Neutron-2 instrument processes signals and data without committing to specific physical components, ensuring functional needs were validated prior to component selection. Finally, a preliminary Physical Architecture was developed using Physical Architecture Block Diagrams (PABs), with Capella's Property Value Management (PVM) add-on and a custom Python4Capella script used to automate mass budgeting and verify compliance with the system mass requirement. Together, these models provide a digital engineering foundation for Neutron-2 that improves requirements traceability, supports early identification of design issues, and reduces reliance on document-centric systems engineering as the project proceeds toward PDR.

CURRENT STATUS AND FUTURE WORK

The team is currently working to complete its Critical Design Review (CDR), which will be held in the Fall of 2026 with the UNP reviewers. A major focus is on producing detailed documentation to support the design. This documentation includes high-fidelity models and analyses that strengthen confidence in the spacecraft's ability to operate in space. In parallel, the team is developing a FlatSat engineering model that will demonstrate an end-to-end mission scenario and

verify that the interfaces, connections, and command paths between components function as intended to collect science data.

Beyond CDR, the team is also preparing for the Flight Selection Review (FSR). The FSR will determine whether the project proceeds to Phase B of the University Nanosat Program and will enable the team to procure flight-quality components and secure a launch. As planning shifts toward the flight model, the team is working to make the FlatSat as representative of the flight system as possible. One way this is being done is by procuring components that are as close to flight-like as possible. For example, the team is preparing two SatNOGS radios for use in two separate flatsats, which will serve as engineering models for the mission. The flatsat will also use a CubeSpace Gen 2 development model, which is intended to function the same way as the flight-model ADCS. In addition, the team will receive a payload from RMD, including the computer and sensor modules, for integration into the flatsat. With these elements combined, the team aims to build a FlatSat that is as close to flight-ready as possible.

CONCLUSION

Gaps in current knowledge of neutron interactions in Low Earth Orbit and in space weather pose risks to both spacecraft and astronauts. The Neutron-2 mission aims to characterize how location and solar activity affect neutron and gamma-ray fluxes to develop an environmental 3D map, improve radiation models, and better inform spacecraft design for radiation exposure. To accomplish this, the mission will conduct three experiments: the first will study the effect of Earth's magnetic field on neutron and gamma-ray fluxes; the second will examine how solar events influence these fluxes; and the third will assess the effect of altitude on neutron and gamma-ray fluxes.

To carry out these experiments, two spacecraft in polar orbits with an altitude difference of at least 50 km will each fly a neutron detector developed by Arizona State University and Radiation Monitoring Devices. The two spacecraft will share the same design, integration approach, testing plan, and operations concept in order to reduce overall mission complexity. They will be launched into separate orbits, and studies are currently underway to evaluate available launch opportunities. As part of UNP NS-12, the mission is currently preparing for CDR, with launch expected no earlier than 2029.

The Neutron-2 mission will share its findings with the scientific community to improve radiation models and strengthen predictions of the environments that space systems may encounter. This knowledge will make

future spacecraft design safer in the future and contribute to the future of human spaceflight programs such as the NASA ARTEMIS program with the return to the Moon.

ACKNOWLEDGEMENTS

The Neutron-2 mission is funded by the University Nanosatellite Program via Contract number CP0089494. The authors gratefully acknowledge contributions from the Neutron-2 student team, HSFL engineering team, UNP review team, Arizona State University, and Radiation Monitoring Devices. In particular, the authors wish to thank the following students from the N2 team: Gianna Longo, Zachary Tamboa, Dennis Sarsozo, Ziar Godoy, Aris Carlos, and Jaycee Hasegawa.

References

1. University Nanosatellite Program, "UNP Nanosatellite 12 User Guide," Rev. A, University Nanosatellite Program, 2025.
2. Singleterry, R. C., F. F. Badavi, J. L. Shinn, F. A. Cucinotta, G. D. Badhwar, M. S. Cloudsley, J. H. Heinbockel, J. W. Wilson, W. Atwell, R. Beaujean, J. Kopp, and G. Reitz, "Estimation of Neutron and Other Radiation Exposure Components in Low Earth Orbit," *Radiation Measurements*, Vol. 33, No. 3, 2001, pp. 355–360.
3. Deiss, H., ed., "Why Space Radiation Matters," NASA, April 13, 2017.
4. Dunai, T. J., *Cosmogenic Nuclides: Principles, Concepts and Applications in the Earth Surface Sciences*, Cambridge University Press, Cambridge, UK, 2010.
5. NASA, "Space Radiation Analysis Group, Johnson Space Center," National Aeronautics and Space Administration, 2024.
6. Nunes, M. A., L. French, P. A. J. Englert, E. J. Pilger, L. Yoneshige, I. Rodrigues, Y. Ben Gershom, A. Imai-Hong, K. Hagi, C. Opie, L. P. Flynn, C. Hardgrove, and B. Roebuck, "Neutron-1 Mission: Low Earth Orbit Neutron Flux Detection and COSMOS Mission Operations Technology Demonstration," *Proceedings of the 33rd Annual AIAA/USU Conference on Small Satellites*, Logan, UT, August 2019.
7. Hardgrove, C., S. T. West, L. E. Heffern, et al., "Development of the Miniature Neutron Spectrometer for the Lunar Polar Hydrogen Mapper Mission," *Proceedings of the 49th Lunar and Planetary Science Conference*, The Woodlands, TX, March 2018, Abstract No. 2341.
8. Hardgrove, C., et al., "LunaH-Map CubeSat," *Proceedings of the Improving Space Operations*

- Workshop*, Pasadena, CA, American Institute of Aeronautics and Astronautics, date unknown.
9. Hardgrove, C., et al., "The Lunar Polar Hydrogen Mapper (LunaH-Map) Mission: Mapping Hydrogen Distributions in Permanently Shadowed Regions of the Moon's South Pole," *Proceedings of the Lunar Exploration Analysis Group Meeting*, Universities Space Research Association, Houston, TX, 2015.
 10. ISISPACE Group, "CubeSat Development Platform Datasheet," Doc. No. ISIS-CDP-DSH-0001, Ver. 1.2, ISISPACE Group, Delft, The Netherlands, 2023.
 11. ISISPACE Group, "ISISPACE Compact EPS 2: ICEPS2 Datasheet," ISIS-ICEPS2-DSH-0001, Version 1.2, Delft, The Netherlands, July 26, 2024.
 12. ISIS—Innovative Solutions In Space B.V., "Antenna Systems," Delft, The Netherlands.
 13. ISIS—Innovative Solutions In Space B.V., "S-band Patch Antenna Datasheet," ISIS.SPPA.DS.00001, Issue 1.4, Delft, The Netherlands, Dec. 10, 2019.
 14. ISIS—Innovative Solutions In Space B.V., "ISIS On Board Computer," Delft, The Netherlands.
 15. CubeSpace Satellite Systems (Pty) Ltd, "*CubeADCS Interface Control Document*," Doc. No. CS-DEV.ICD.CA-01, Ver. 1.05, Stellenbosch, South Africa, Feb. 20, 2026.
 16. Hawai'i Space Flight Laboratory, "ADCS Test Facility," University of Hawai'i at Mānoa, Honolulu, HI, accessed 12 June 2026. <https://www.hsfl.hawaii.edu/adcs>
 17. Libre Space Foundation, "SatNOGS COMMS Interface Control Document," Version 1.6. From <https://gitlab.com/librespacefoundation/satnogs-comms>
 18. Jet Propulsion Laboratory, "F' User Manual," F Prime Documentation, Ver. 3.6.2, California Institute of Technology, Pasadena, CA, accessed 12 June 2026.
 19. Space Weather Live, "*Real-Time Solar Activity*," Space Weather Live, <https://www.spaceweatherlive.com/en/solar-activity.htmlv>
 20. National Aeronautics and Space Administration, "Solar Dynamics Observatory (SDO)," NASA Goddard Space Flight Center, Greenbelt, MD, accessed 12 June 2026. <https://sdo.gsfc.nasa.gov/>
 21. Piper L. Kline, William Edmonson, Miguel A. Nunes. "Developing a MBSE Methodology for CubeSats" 2025. Small Satellite Conference, SSC25-P1-78.
 22. Piper Kline. "Developing A Model-Based System Engineering Methodology For Cubesat

Missions" M.Sc. Thesis Published December 2025.